

Titanic Dataset - Exploratory Data Analysis (EDA)

Objective

Perform Exploratory Data Analysis (EDA) on the Titanic dataset to uncover hidden insights, trends, and patterns that influenced passenger survival.

Dataset

Source: Kaggle Titanic Dataset

Features: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

EDA Process

- Data Loading & Exploration: Used `info()`, `describe()`, `value_counts()` to understand data types and distributions.
- Data Cleaning: Filled missing values in Age and Embarked, dropped Cabin and irrelevant columns.
- Univariate Analysis: Explored Survived, Sex, Pclass, Age.
- Bivariate & Multivariate Analysis: Used plots to examine relationships.
- Survival Summary: Counted Survived/Dead by Sex and Pclass.

Key Insights

Survival Summary by Gender and Class:

Sex	Pclass	Survived	Dead
Female	1	91	3
Female	2	70	6
Female	3	72	72
Male	1	45	77
Male	2	17	91
Male	3	47	300